Complex Systems, Costly Change: How Functional Feedback Shapes Policy Punctuations

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Punctuated Equilibrium Theory (PET) explains policy change as long periods of stability punctuated by brief bursts of major change, but its empirical focus has largely been on decision costs—especially budgetary ones—while overlooking other constraints. This paper argues that functional costs—the drafting, harmonization, and compliance burdens inherent in producing legislation—act as dynamic feedback shaping when and how punctuations occur. I develop a dynamic stochastic model, grounded in complex systems theory, in which functional costs evolve with the size and interconnectedness of a policy subsystem. Simulations show that these costs can dampen extreme changes without external shocks, generating punctuated patterns endogenously. I test this mechanism using 69,000 EU legislative acts (1980–2015), operationalizing functional costs as the scaled average degree of each policy field's legal citation network. Quantile regressions reveal that higher functional costs significantly reduce the magnitude of changes at the upper tail of the distribution, with effects varying by legal instrument: regulations are less sensitive to costs than decisions, consistent with greater centralization and modularity in their drafting. These findings reframe PET's cost mechanism, highlighting subsystem complexity as a structural driver of policy stability and change.

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1 Introduction

Policy change is a complex and dynamic phenomenon. Understanding what triggers or inhibits change is crucial for policy-makers, researchers, and practitioners who seek to address societal problems and improve policy outcomes. Consequently, various theoretical frameworks and models attempt to explain how different factors can interact and influence policy change over time. Some of these frameworks focus on specific aspects, such as institutions (North 1990a; Pierson 2000; Thelen 2004), interests (Sabatier 1988; Kingdon 1995), ideas (Hall 1993; Béland 2009; Schmidt 2010), networks (Rhodes 1997; Klijn, Steijn, and Edelenbos 2010; Cairney 2016), or events (Birkland 2006). Others try integrating multiple factors into a comprehensive approach, such as punctuated equilibrium theory (Baumgartner and Jones 1993).

At its core, PET describes systems that produce a pattern of change that exhibits long periods of stasis occasionally interrupted by brief periods of meaningful and substantial change. Baumgartner and Jones (1993) adopted the theory stemming from paleobiology (Gould and Eldredge 1977) to describe the patterns observable in the change of public agendas. While initially focused on explaining stability and change, PET has been extended to a general information processing theory (Jones and Baumgartner 2004). The theory suggests that policy systems are confronted with policy inputs that consist of nearly infinite components. These inputs can be assumed to sum to an approximately normal distribution (see also Padgett 1980). Although this assumption has been challenged (Desmarais 2019), staying with it, a perfect responsive system should produce normally distributed outputs from normally distributed inputs.

Friction and positive or negative feedback are proposed mechanisms for why this is not the case (Jones and Baumgartner 2004). Yet, while the effects of friction have been extensively studied (e.g., Baumgartner et al. 2009), more general feedback mechanisms have been underexplored. In theory, positive feedback mechanisms amplify signals given by policy inputs, leading to punctuations, while negative feedback mechanisms inhibit the response to inputs, causing incremental change patterns (Jones and Baumgartner 2004). Although a central mechanism within the explanation of punctuated patterns, most empirical studies do not directly engage with these feedback mechanisms attributed to creating punctuations (Kuhlmann and van der Heijden 2018).

This paper argues that the focus on decision costs, primarily tied to budgetary changes, has overshadowed other critical factors like the costs of drafting, harmonization, and compliance, particularly in the context of legislative outputs in complex settings. This has constrained theoretical contributions and empirical investigations. Furthermore, while PET aligns with complex systems theory (Jones, Sulkin, and Larsen 2003; Jones and Baumgartner 2012), few studies have probed how feedback mechanisms from complex systems influence policy change (Stauffer et al. 2022), leaving a gap in understanding the connection between policy inputs and outputs (Desmarais 2019) and the role of inter-subsystem dynamics (Jochim and May 2010).

In response to these gaps, this paper introduces a theoretical model and provides empirical evidence to demonstrate how dynamics across policy subsystems generate feedback mechanisms that accelerate or inhibit policy punctuations. By introducing the concept of dynamic feedback, derived from complex systems theory, it offers an additional explanation for punctuations in policy outputs.

Dynamic feedback recognizes that policy-makers face temporally bounded challenges when producing policy outputs that exhibit different dynamics over time. It highlights that functional constraints alone can lead to punctuated output without external influences such as crisis events. To test the theoretical mechanism developed, I analyze the change distributions of 35 years of policy outputs from the European Union between 1980 and 2015. The mechanism is tested using quantile regression, with measures retrieved from a network analysis of the legal landscape to capture functional costs policy-makers face. The results indicate that higher functional costs in a given year decrease the likelihood of punctuation occurring in subsequent periods.

The paper is structured as follows: Section 2 discusses the role of costs in punctuated equilibrium theory (PET), focusing on functional costs. Section 3 introduces the theoretical model and demonstrates how it can simulate policy-making under information constraints. Section 5 outlines the empirical strategy used to identify the effect of functional costs on policy punctuations in EU policy-making. Section 6 presents the results, and Section 8 concludes the study by summarizing the findings and discussing their implications.

2 Costs in policy making

When examining patterns of change, the literature on policy punctuations commonly emphasizes the role of political institutions and the extent to which these institutions encounter costs that may create obstacles in the policy-making process (Robinson et al. 2007; Baumgartner et al. 2009; Jones et al. 2009; Breunig 2011). Jones, Sulkin, and Larsen (2003) categorize the costs faced by decision-making systems into four types: decision, transaction, information, and cognitive costs.

Linking to Buchanan and Tullock (1965), they define decision costs as costs that occur for reaching an agreement across political actors. While they separate transaction and informational costs, I argue in line with North (1990b) that they are highly intertwined. They state: "The costliness of information is the key to the costs of transacting, which consist of the costs of measuring the valuable attributes of what is being exchanged and the costs of protecting rights and policing and enforcing agreements." (p. 27). Regarding inhibiting change, one can think of these costs as costs that arise due to the functional nature of policy systems. Hence, in this paper, I will refer to them as *functional costs*. Lastly, they define cognitive costs as stemming from the limited processing capacity inherent in any social institution that is comprised of human individuals. This reflects the bounded rationality approach adopted to PET (see also Simon 1957; Padgett 1980).

In terms of relevancy for them, "[...] costs of bringing relevant parties to agreement in a system of separated powers (decision costs) generally outweigh the costs of holding hearings, enacting statutes, or changing budgetary allocations once agreement has been reached (transaction costs)." (p. 154). This line of argument has two caveats.

First, Jones, Sulkin, and Larsen (2003) rest their argument on the institutional level. Hence, for them, the decision-making procedure of the institution and, most importantly, the *differences* in decision-making procedures across institutions are the primary source of cost differences that might explain differences in punctuated patterns (see also Workman, Jones, and Jochim 2009). Yet, this distinction has no explanatory power when the focus is on dissecting *within* variance of institutional outputs and explaining individual punctuations, especially regarding legislative outputs. Here, we can assume that cognitive and even decision costs stemming from the institutional setup are relatively stable (Mortensen 2009). Although Spiller and Tiller (1997) show that decision costs under certain conditions can be manipulated as well.

The latter part of the statement highlights a secondary limitation: the assumption that functional costs only materialize after an agreement has been reached does not fully account for their presence at multiple stages throughout the policy-making process (Krutilla 2011). Scholars such as Schuck (1992) argue that functional costs are a persistent feature at various stages of policy development. According to them, a law with higher associated costs

"[...] tends to be more costly and cumbersome to administer, more difficult for lawmakers to formulate and agree upon, and more difficult to reform once established" (p. 18). Consequently, Hurka and Haag (2020) demonstrate that the length of legislative processes can be significantly extended with rising costs, while Haag, Hurka, and Kaplaner (2024) find that the success of policy implementation is heavily dependent on them.

Therefore, functional costs can be understood as reflecting the resources required to navigate the complexities of the existing legal and regulatory framework when developing new policies. This includes costs associated with drafting legislation, ensuring consistency with existing laws, harmonizing regulations across different sectors or jurisdictions, and even implementing and enforcing new policies.

Functional costs arise from the functional nature of policy systems — they are inherent in the processes and structures required to develop and implement policies effectively. These costs are dynamic and can fluctuate based on the complexity of the policy environment, the interdependencies among existing regulations, and the administrative capacity of the institutions involved (see also Simon 1962; Pierson 2000).

This perspective challenges the traditional focus of early policy punctuation research on decision costs, which is primarily derived from a focus on budgetary changes and their relatively straightforward transmission. However, when it comes to legislative decisions, the reality is considerably more complex. These decisions often represent just the starting point or even a repeated process, with subsequent requirements to ensure compliance with the existing legal framework — a costly process making it far from trivial.

To illustrate, consider the process of drafting new financial regulations aimed at preventing systemic risk in the banking sector. Policy-makers must navigate a web of existing national and international economic rules, balance central bank policies, and align with the Basel III framework in the case of the EU. Functional costs in this case include:

- Extensive legal research to ensure compatibility with current financial laws.
- The involvement of multiple regulatory bodies to harmonize banking standards across borders.
- Consultations with industry experts to avoid unintended disruptions in capital flows.

These functional costs differ from political costs (e.g., gaining support from various interest groups) and cognitive costs (e.g., understanding the intricacies of complex financial instruments). They represent the operational effort required to develop policies that can fit within the intricate and highly interconnected regulatory environment governing global finance.

Figure 1 illustrates these complexities. It shows the legal landscape of economic policies in the European Union in 2015, with each node representing a legislative output and each edge representing references to other legislative outputs. This shows that the legal landscape is often highly complex and interdependent. Consequently, transmitting decisions into this legal framework and or designing policies that fit into it in the first place is very costly. It has the potential to affect the policy-making process significantly. Furthermore, every policy added to the system has the potential to influence the number of functional costs for future policy-making endeavors (see Pierson 2004), making the system highly dynamic. This highlights the need for a more thorough understanding of functional costs, recognizing them as integral and ongoing throughout the policy cycle.

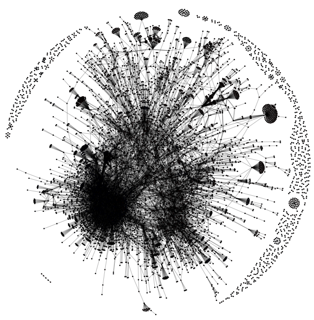

Figure 1: Legislative network of economic policies in the EU in the year 2015. Data from Fjelstul (2019)

3 Models of policy change

The strong emphasis on decision costs has also permeated models that describe how systems produce punctuated outputs. Specifically, the model by Jones and Baumgartner (2004) considers costs as a form of friction. This friction functions as a stick-slip mechanism, influencing the system's dynamics.

Their model takes an input, denoted as S_t, from a standard normal distribution and produces a response, denoted as R_t.

The transformation of inputs into responses hinges on three parameters[1]:

[1]The original paper uses the denominator C for the friction parameter. To avoid confusion with the functional cost parameter further down, i will refer to it as F.

$$F = \text{friction}$$

$$\lambda = \text{efficiency}$$

$$\beta = \text{amplification}$$

The response is given by R_t:

$$R_t = \begin{cases} \beta S_t & \text{if } F < (S_t + \sum S_{0<k}) \\ \lambda\beta S_t & \text{else} \end{cases} \tag{1}$$

The dynamic aspect of the model pertains only to the $S_t + \sum S_{0<k}$ mechanism. Essentially, in their model, inputs are only partially fulfilled if they do not cross the threshold of the stable friction parameter F since they are multiplied by the efficiency parameter λ that is bound between 0 and 1 ($0 < \lambda < 1$).

$\sum S_{0<k}$ represents the sum of unfulfilled input signals from the prior periods and is added to the current input signal S_t.

Consequently, their model suggests that inputs are only partially realized unless they surpass the friction threshold F, being then multiplied by the efficiency parameter λ (where $0 < \lambda < 1$). When the sum of inputs exceeds the friction threshold, inputs are either completely fulfilled or amplified by the parameter β, leading to punctuated outputs.

This interpretation of friction aligns closely with the emphasis on decision costs (Workman, Jones, and Jochim 2009). Indeed, if decision costs represent the predominant costs, one might envision them functioning in a stick-slip manner. Once a decision is reached and the friction is fully overcome, punctuations are likely to occur, signaling abrupt shifts in output or policy.

3.1 Limitations of previous models

However, the model exhibits significant shortcomings in explaining how functional costs impact policy punctuations. First, it is improbable that these costs behave like friction. The major difference is that they can not be overcome suddenly since they are functional in nature. For example, policy-makers can not just ignore other provisions in the legal landscape when drafting new ones. Instead, I propose that they act as negative feedback on policy demands, moderating or restraining changes rather than facilitating and/or amplifying them (Pierson 2000).

Second, the model focuses primarily on the institutional level, limiting its capacity to account for how cross-sectoral dynamics within institutions could influence policy-making patterns (Jochim and May 2010). This limitation makes applying the model to more diverse policy environments hard, where policy-makers have certain flexibility to choose when and where to react to policy demands. Lastly, while the model introduces a dynamic input mechanism, it treats costs as static. This assumption needs to be revised, particularly concerning dynamic functional costs, which can fluctuate based on numerous factors. A more realistic assumption would involve a dynamic integration of costs, reflecting their varying impacts over time.

3.2 A dynamic stochastic model of policy outcomes under information constraints

Therefore, I propose to focus on a dynamic stochastic model that accounts for functional costs and their cross-sectoral variance and models the decision-making as a dynamic optimization process where policy-makers try to optimize the policy outputs across multiple sectors by incorporating considerations based on demand and costs (see also Bendor, Moe, and Shotts 2001). This model aligns with complex systems theory, where policy-making systems are viewed as adaptive. Functional costs serve as feedback mechanisms, creating emergent patterns in policy outputs that reflect complex systems' non-linear and interdependent nature. I show in the following that this model can explain punctuated outputs without friction parameters.

The model contains the following main parameters:

$$C = Cost$$

$$D = Demand$$

$$U = Utility$$

Each of these parameters varies across time (t) and field (i). Additionally, there is a set of parameters which I will call modifiers:

$$x = scale$$

$$\phi_{t,i} = coverage$$

Similar to prior models, demand for a given field at a given time point ($D_{t,i}$) is defined by demand drawn from a normal distribution for each period plus the unfulfilled demands from the prior period ($D_{t-1,i}$). The unfulfilled demand represents the demand from the previous period minus the sum of fulfilled policy outputs ($\sum P_{t-1,j}$) which gets multiplied with a coverage parameter ϕ. ϕ defines the extent to which the demand was met in previous periods.

$$D_{t,i} = N(0, x) + (D_{t-1,i} - \sum P_{t-1,i,j} * \phi_{t-1,i}) \tag{2}$$

The initial functional cost ($C_{0,i}$) of a field is drawn from a normal function with the standard deviation of $\frac{1}{x}$. The scale parameter x captures the relationship in scale between costs and demand. I consider functional costs to be strictly positive; thus, the model takes the absolute value of the draw. Therefore, contrary to institutional setups or the like, which can act in both directions, enabling or hindering change in a given field, functional costs are initially strictly positive in the sense that costs always exist and inhibit the potential to create policies.

$$C_{0,i} = |N(0, \frac{1}{x})| \tag{3}$$

To calculate the costs for a specific time and domain ($C_{t,i}$), the model incorporates both initial costs and the costs associated with adding policies to the policy stock. When policy-makers opt to expand the policy portfolio, they inherently raise the cost for the domain, as they must accommodate the impact of each realized policy when considering future policy decisions. Consequently, each new policy can escalate forthcoming costs (Schuck 1992). Although theoretically policy-makers could opt to dismantle current policies, I operate under the assumption, supported by Adam et al. (2019), that policy-makers are strongly incentivized to accumulate new policies rather than dismantle existing ones.

In the model, the added cost is variable and results from multiplying a draw from the same normal function as before with the number of policy items realized in the previous period. Individual policies are given by the binary variable $P_{t,i,j}$. Essentially, $P_{t,i,j}$ denotes if a particular policy item was realized (= 1) or was not realized (= 0). Therefore, $\sum P_{t-1,i,j}$ denotes the number of realized policies in a given field for the previous period.

$$C_{t,i} = C_{t-1,i} + |N(0, \frac{1}{x})| * \sum P_{t-1,i,j} \tag{4}$$

Lastly, the utility of realizing a policy $U_{t,i,j}$ in a given field for a given time is calculated by taking the difference between the demand and costs. This captures the core thought that policies that are cheap to implement but have strong demand have more utility for the policy-makers than costly policies without much demand attached. Additionally, $\phi_{t,i,j}$ represents the coverage parameter that denotes how much of the demand was fulfilled by the implemented policies.

$$U_{t,i,j} = D_{t,i} - C_{t,i} - \sum P_{t,i,j} * \phi_{t,i,j} \tag{5}$$

How policy-makers optimize given these conditions is essential to the question of whether this setup creates punctuated output distributions. For this, I will consider a limited capacity model. Under this assumption, policy-makers have a limited number of policies they can output. The total number of policies fulfilled across all fields per period is bounded between 0 and the system's capacity[2].

$$0 <= \sum P_{t,i,j} <= capacity \tag{6}$$

Given this limitation, policy-makers try to find the optimal solution (ω_t) by considering different utility sets. If policy-makers would have complete information about the expected values, the solution to the problem would be straightforward:

$$\omega_t = max \sum U_{t,i,j} \tag{7}$$

The primary challenge for policy-makers within this framework stems from the incomplete information available to them and, consequently, their bounded rationality (Simon 1957; Padgett 1980). Specifically, while policy-makers might have a clear understanding of the

[2]In the following, I define capacity as the number of policies a system can produce within a given period. Although I do not explicitly model how more costly policies impact this capacity, the assumption is implicitly incorporated through their lower utility, as costlier policies are less likely to be selected. For further details, see Section 4.

demand for a new policy ($D_{t,i}$) and its associated costs ($C_{t,i}$) since they are aware of the policy environment in place at the time, they lack comprehensive details regarding the utility ($U_{t,i,j}$) that each policy would yield. This is because the utility's calculation hinges on the actual outcomes of policy implementation, represented by the product of policy output ($P_{t,i,j}$) and its coverage parameter ($\phi_{t,i,j}$), to determine if the policy demand has been adequately met.

In essence, while policy-makers are informed about the costs and demand for new policies, they remain uncertain about how much a specific policy will fulfill the existing demand. As a result, they are constrained to optimizing the expected total utility ($\sum \hat{U_{t,i,j}}$), reflecting their best attempt to address policy demands under the limitation of partial information.

$$\hat{U_{t,i,j}} = P_{t,i,j} * (D_{t,i} - C_{t,i}) + \epsilon \tag{8}$$

Unlike simpler models that rely on static parameters (like friction), this approach emphasizes the dynamic optimization process of policy-makers. They must continuously adapt their strategies based on changing demands and the accumulated costs of past decisions, reflecting the inherently adaptive and feedback-driven nature of complex systems. Moreover, the integration of functional costs as dynamic variables highlights how constraints are not merely static barriers but are actively influenced by prior outputs and demands in an ongoing manner. The model's structure, allowing for fluctuating costs based on past actions, ensures that these costs act as evolving feedback rather than fixed friction points. This dynamic interplay mirrors real-world policy environments, where interdependencies and the effects of prior decisions influence current and future policy choices, thus embodying the essence of complex systems theory.

4 Simulation

I use a simulation approach to test if punctuated patterns emerge given the model above. As highlighted by Sterman (2000), the benefit of simulations is that they allow us to model dynamic systems where analytical solutions may not be feasible due to the complexity and interdependence of variables involved. Since punctuations are inherently complex patterns arising from the interplay of feedback loops, functional costs, and varying demands across policy fields, solving the model analytically would require overly simplifying assumptions that may not capture the system's full dynamics. On the other hand, simulations provide a flexible approach to explore the decision process under varying conditions, allowing the model to operate with all its constraints and interdependencies intact. This method enables me to investigate how policy outputs vary over time and under the constraints outlined above, revealing the precise mechanisms that produce punctuations or periods of stability without reducing the model's complexity.

4.1 Assumptions

So, how do policy-makers choose policy outputs under these settings? What expectations can we derive from the model above?

First, I must state the assumption under which I simulate the decision-making. The assumptions are as follows:

- **(A1) Coverage Knowledge Post-Implementation:** The value of $\phi_{t,i,j}$ is only known after policies have been realized (Simon 1957; Padgett 1980).
- **(A2) Continuous Policy Production:** Policy makers are inclined to constantly produce policy (Adam et al. 2019).
- **(A3) Costly and Limited Information Sampling:** $\sum \hat{U_{t,I,j}}$ can only be gauged when observed; this process is costly and, hence, limited.
- **(A4) Capacity Constraints:** Capacity to output policy is limited but used fully (Baumgartner and Jones 1993).

Therefore, policy-makers try to fill up a constant amount of policy output activity across the policy fields. Furthermore, I assume that the total value $\sum \hat{U_{t,i,j}}$ can only be gauged by policy-makers by sampling different sets of $\hat{U_{t,i,j}}$ and is not a priori known. Since this process is costly, policy-makers can only inspect a finite set of samples before deciding on the set of ω_t that optimizes $\sum \hat{U_{t,i,j}}$. This constraint limits the possibility of policy-makers focusing solely on one field, as they cannot exhaustively evaluate every possible scenario or concentrate all resources on a single domain. Instead, they must operate under bounded rationality, diversifying their attention across multiple fields to maximize overall utility within the constraints of limited information and evaluation costs.

Nevertheless, given the optimization process and the role of costs in determining utility, fields with lower costs inherently have a higher likelihood of being selected. When faced with similar demand levels across fields, policy-makers are more inclined to focus on those with lower associated costs, as these provide higher utility and are less resource-intensive to implement. This tendency aligns with their objective to optimize outputs efficiently while minimizing expenditure, ensuring that fields with manageable costs and high utility are prioritized within the constraints of the sampling process. I integrate this decision approach by allowing policy-makers to construct possible solutions repeatedly and gauging their expected utility before deciding to realize a set of policies.

What patterns can be expected under these assumptions? Given the fixed number of policies, I anticipate high demand and relatively low costs during this period, resulting in a greater allocation of policies to a field. In turn, this allocation has the potential to raise the field's costs in the following periods, decreasing the chance of high allocations. This dynamic should result in a punctuation pattern, with few periods of high allocation and many periods of stasis while other fields catch up in costs.

4.2 Scenario and results

The scenario contains $i = 1, 2, ..., 20$ policy fields and spans a period of $t = 1, ..., 30$. For the demand and cost development, I set $x = 10$. The coverage parameter is defined as $\phi_{i,t} = U(0, 1)$. Capacity is limited to an output of 100 policies per period and policy makers sample 100 different utility sets by drawing random combinations of policies from different fields before deciding on the highest utility one.

Based on the assumptions, I define the simulation function as presented in Figure 2.

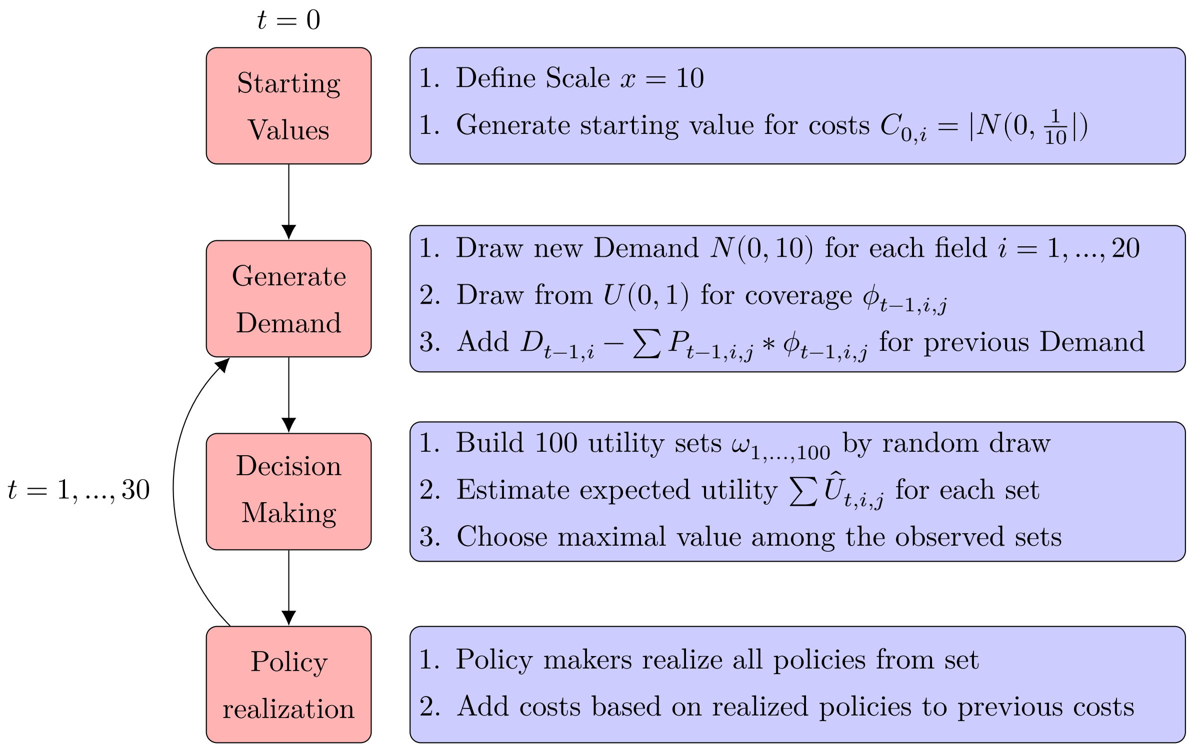

Figure 2: Simulation pseudo code.

The simulation generates a dataset that includes the number of policies fulfilled annually within each field and the corresponding functional costs for each period. Utilizing this, I compute the percentage changes in output for each field and period. Running the model under those conditions produces punctuated outputs, as seen in the Supplementary Information (SI). I employ quantile regression to visually represent functional costs' influence on policy punctuations. This method enables me to predict the impact of functional costs across various quantiles (τ) of percentage changes. If functional costs influence policy punctuations, I anticipate observing a negative coefficient of functional costs for high values of τ (see also Breunig and Jones 2011). To ensure representative results, I repeat this procedure 100 times. Figure 3 showcases the resulting coefficients derived from these 100 simulations.

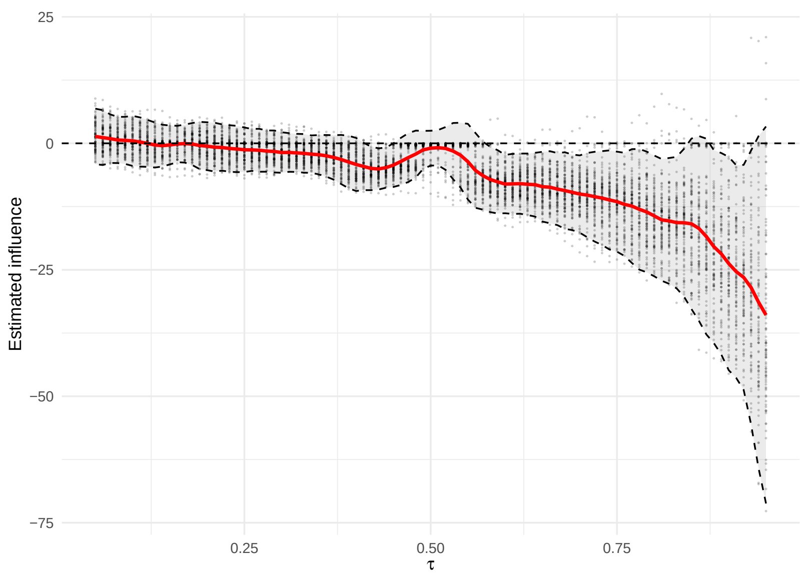

Figure 3: Results of quantile regression based on simulated scenarios. The simulation was conducted 100 times. The dependent variable are percentage changes per field and period; the independent variable is period standardized costs per field similar to the specification outlined in Equation 12. The quantile regression includes a time trend control to capture demand dynamics, similar to yearly fixed effects as in Equation 12. Each dot represents one coefficient estimate from a simulated scenario. The main line represents the mean coefficient estimate across all scenarios. The upper and lower lines mark the 95% confidence interval.

It becomes evident that higher quantile changes are inversely correlated with the functional cost factor under this model. The connection is especially prevalent in higher quantiles, thus resulting in more extreme policy changes. This suggests that greater costs in a field during a specific time period lead to less severe policy punctuations in the model. Conversely, lower relative costs exacerbate the severity of policy shifts. This can be summarized into one expectations for the empirical setting:

- E: The higher|lower the (relative) subsystem functional costs in a given period, the less|more pronounced are policy punctuations

5 Empirical strategy

The model shows that under certain circumstances functional costs can influence the occurrence of policy punctuations. Yet, the question remains if this pattern also manifests in reality. To answer this question I will use legislative production of the European Union to test the expectations formulated above.

The European Union (EU) offers an especially suitable setting for testing the theoretical argument on policy punctuations and dynamic feedback mechanisms. Its complex legal and institutional structure combines diversity across policy fields with variation in decision-making processes, creating conditions to observe how sectoral differences shape these dynamics. Unlike national governments, the EU enjoys relative autonomy from direct public pressure, enabling the analysis of functional costs and feedback processes in a setting less confounded by short-term opinion shifts (Princen 2013).

This suitability is reinforced by the EU's extensive and well-documented legislative record, with consistent data from 1980 to 2015 across numerous policy domains. The EuroVoc classification system groups acts into distinct fields (Kaplaner, Knill, and Steinebach 2025), while the EU's densely interconnected legal landscape—mapped through citations and references—supports network analysis to quantify subsystem costs (Fjelstul 2019), directly linking the empirical setting to the theoretical framework.

5.1 Dependent Variable

The dependent variable in this analysis is the *yearly percentage change in policy outputs* across the European Union (EU) from 1980 to 2015, captured based on 69,483 policy outputs distributed across 16 distinct fields. The outputs are taken from the Fjelstul (2019) dataset and are limited to secondary legislation. Additionally, I employ the EuroVoc classification system to structure policy outputs into distinct fields (see also Kaplaner, Knill, and Steinebach 2025). EuroVoc provides a standardized and comprehensive categorization of EU legislative acts, allowing for consistent measurement and field-specific comparisons. By using this system, the analysis ensures that sectoral distinctions are consistently applied. Measuring the yearly percentage changes captures the relative fluctuations in policy activity and allows for the identification of punctuations as deviations from periods of stasis or incremental change.

The calculation for percentage change follows the standard approach:

$$Pct.change_{i,l,t} = \frac{N_{i,l,t} - N_{i,l,t-1}}{N_{i,l,t-1}} * 100 \tag{9}$$

Where:

- $N_{i,l,t}$ is the number of policy outputs in field i, legislative instrument l, at time t

This operationalization captures the intensity of change within each field, providing a dynamic view of legislative activity. The approach normalizes variations by sing percentage changes rather than absolute counts, allowing for cross-sectoral comparisons over time.

Lastly, as noted in Equation 9 the percentage changes are calculated separately for each legal instrument. As previously mentioned, the logic behind the choice of instrument varies significantly (Hurka and Steinebach 2021), affecting the amount of functional costs incurred by the EU. Disaggregating the percentage changes by legal instrument allows testing of these effects. The observation period starts in 1980; however, the first observation year is dropped due to its use in calculating the percentage changes. Additionally, because a lagged version of this variable is used as a control (see Section 5.4), observations from 1981 are also excluded from the analysis. Finally, since not all periods across all fields and instruments had legislative output, I exclude periods that follow a period with zero output to avoid calculations resulting in infinite values. For a detailed overview of the number of included periods per policy field, please refer to the supplementary information (SI).

5.2 Independent variable

I operationalize functional costs through the average degree of the legal citation network within a given field at a given time. This approach is grounded in network theory, where the degree of a node indicates the number of connections it has with other nodes. In the context of legislative outputs, the degree represents the extent to which a given policy item cites or is cited by other policy items. The average degree, in turn, captures, on average, how interconnected the nodes within the network are.

This metric reflects the complexity and interconnectedness (see Bonchev and Buck 2005) of the legal framework within each field and serves as a proxy for the subsystem's functional costs. The reasoning is that the more interconnected and interdependent the legislative items are, the higher the cost for policy-makers to create, adapt, or implement new policies. Each additional connection implies that any new policy must consider, align with, or adjust to a more significant number of existing regulations, thereby increasing the drafting, harmonization, and compliance efforts. These efforts translate into higher functional costs, as policy-makers need to ensure consistency, address conflicts, and integrate new measures into an increasingly complex and interconnected legislative environment (Burgin and Mestdagh 2020; Koniaris, Anagnostopoulos, and Vassiliou 2018).

Additionally, a higher average degree signifies a dense regulatory network, suggesting that changes in one part of the subsystem could lead to ripple effects throughout the network, further complicating the policy-making process. This interconnectedness requires additional resources, expertise, and time, further raising the functional costs of policy-making. The average degree encapsulates the effort and resources needed to navigate and modify the subsystem's legal landscape, reflecting how costly it is to implement changes that align with the existing regulatory framework (Kütt and Kask 2021; Burgin and Mestdagh 2020).

Similar approaches have been used by other scholars either to (partially) s the complexity of individual provisions (Katz and Bommarito II 2014; Hurka, Haag, and Kaplaner 2022; Senninger 2023) or to s the complexity of the legal landscape as a whole (Lee, Lee, and Yang 2019). The average degree $A_{i,t}$ for a field i at time t is defined as:

$$A_{i,t} = \frac{\sum_{j}^{N_{t,i}} k_{j,i,t}}{N_{t,i}} \tag{10}$$

Here, $N_{t,i}$ represents the total number of policy items in field i at time t, while $k_{j,i,t}$ denotes the degree of policy item j. Again, I use the EuroVoc classification for field identification. Additionally, I include connections where at least one node belongs to the specified field. The data on the connections between legal documents is taken from Fjelstul (2019). Only active and legally binding policy items are included for each time period to ensure that the measure accurately reflects the current functional environment. This information is taken from Borrett and Laurer (2020). This ensures the analysis focuses on the actual subsystem costs impacting policy-making within each period rather than including outdated or repealed documents.

Figure 4: Avg. Degree over time across all policy fields.

Yet, some caveats arise when using this measure. As shown in Figure 4, the average degree of all policy fields exhibits an upward trend over time (see also Hurka, Haag, and Kaplaner 2022). This trend suggests that using the average degree directly may conflate the impact of rising costs with a general time trend rather than isolating the specific effect of functional costs in the legislative process.

I use a scaled version of the average degree (SAD) to address this issue. According to the theoretical model, policy-makers operate within specific time frames when deciding on policy outputs, making a field's *relative* costs more important than absolute values. Therefore, to better align the measure with this theoretical perspective and eliminate the risk of capturing mere temporal trends, I scale the average degree relative to the costs observed in other policy fields within the same period.

This adjustment achieves two main objectives: first, it removes the time trend from the data, as demonstrated in Figure 4, ensuring that the measure focuses on cross-sectional variance rather than temporal patterns. Second, it aligns with the theory that policy-makers prioritize relative costs, making the measure more theoretically consistent and empirically robust.

The final measure can be expressed as:

$$SAD_{i,t} = \text{Scaled Avg. Degree}_{i,t} = \frac{A_{i,t} - \mu_t(A)}{\sigma_t(A)} \tag{11}$$

Where $A_{i,t}$ is the average degree for field i, and μ_t is the mean of all policy fields within the same period with σ_t representing the standard deviation. This formulation ensures that the measure reflects the relative costs within the policy landscape while accounting for temporal dynamics.

Additionally, in the models outlined in Section 5.4, the variable is lagged by one period (SAD_{t-1}) to account for the fact that policy-makers react to the information available from the previous period when making decisions. This lagged structure reflects the temporal nature of decision-making processes. It aligns with the theoretical model, which assumes policy-makers respond to the latest data rather than contemporaneous conditions.

5.3 Controls

The empirical specification includes several controls to refine the analysis. First, yearly fixed effects account for time-specific factors (e.g., economic cycles or political shifts), isolating the relationship between functional costs and policy changes from broader temporal fluctuations. Field fixed effects capture differences across policy areas, ensuring that comparisons of functional costs are made within each field's unique context. Controls for legal instruments (Regulations, Directives, Decisions) adjust for their distinct administrative implications. Lastly, a lagged percentage change accounts for temporal dependencies in policy activity, helping to capture autocorrelations and reduce the influence of past fluctuations on current outcomes (see also Holyoke and Brown 2019).

5.4 Model specification

The following quantile regression model is used to estimate the conditional quantiles of the percentage change in policy outputs (one observation per field i × instrument l × year t):

$$
\begin{aligned}
Q_\tau(\text{Pct. change}_{i,l,t} \mid X_{i,l,t}) = \alpha_\tau &+ \beta_{1,\tau}\,\text{Pct. change}_{i,l,t-1} + \beta_{2,\tau}\,\text{SAD}_{i,t-1} \\
&+ \gamma_\tau^{\text{Regulation}}\,\mathbf{1}\{\,l = \text{Regulation}\,\} + \gamma_\tau^{\text{Directive}}\,\mathbf{1}\{\,l = \text{Directive}\,\} \\
&+ \lambda_{i,\tau} + \mu_{t,\tau}.
\end{aligned}
\tag{12}
$$

Where:

- $\text{Pct. change}_{i,l,t}$ denotes the percentage change in policy outputs for field i, instrument l, and year t.
- $\text{Pct. change}_{i,l,t-1}$ is the lagged percentage change (within the same field–instrument cell).
- $\text{SAD}_{i,t-1}$ is the lagged scaled average degree (functional cost proxy) for field i.
- $\mathbf{1}\{\,l = \text{Regulation}\,\}$ and $\mathbf{1}\{\,l = \text{Directive}\,\}$ are instrument dummies; **Decision** is the omitted baseline.
- $\lambda_{i,\tau}$ are field fixed effects; $\mu_{t,\tau}$ are year fixed effects.
- $X_{i,l,t}$ collects all regressors above.

The model is estimated using *conditional quantile regression*, which ses how regressors influence different points τ of the conditional distribution of $\text{Pct. change}_{i,l,t}$ (Koenker and

Bassett 1978). The coefficient $\beta_{2,\tau}$ captures the association between higher relative subsystem costs ($\text{SAD}_{i,t-1}$) and the τ-th conditional quantile of percentage changes. A negative $\beta_{2,\tau}$ indicates that higher costs in the previous period are associated with smaller percentage changes at quantile τ; a positive value indicates the opposite. This is especially informative at high τ (e.g., 0.90, 0.95), where punctuated changes reside (see Breunig 2011; Breunig and Jones 2011).

6 Results

6.1 Results from quantile regression

The results of the quantile regression are visualized in Figure 5, displaying the coefficient estimates from the specification in Equation 12 alongside their respective 95 percent confidence intervals. I will focus on the upper end of the distribution, specifically the 90th and 95th quantiles, as they capture cases of significant policy change. At this point, it is essential to note that overall, the patterns of the change distributions follow a punctuated pattern as assessed in the SI.

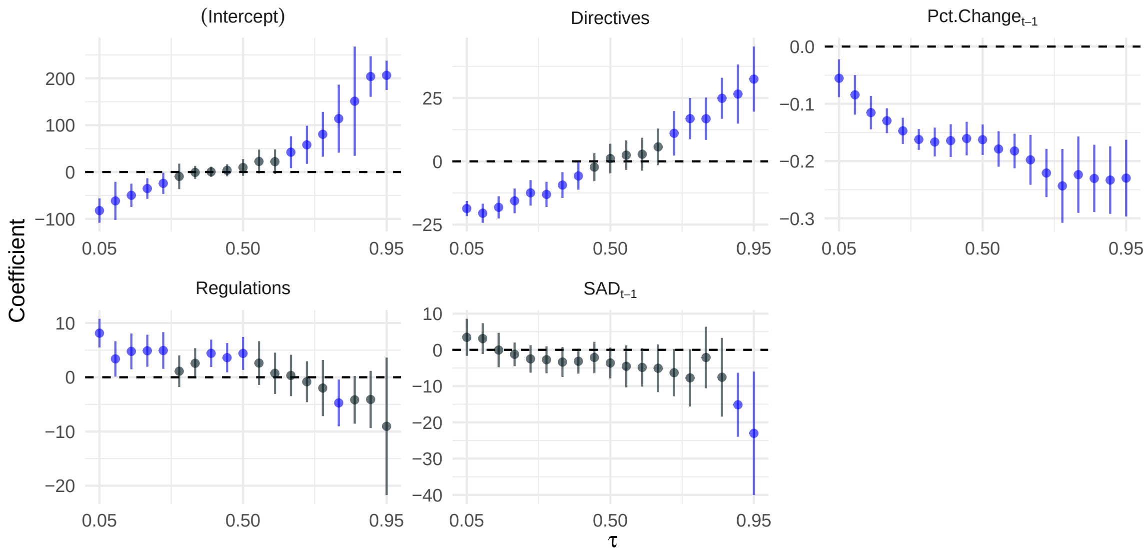

Figure 5: Coefficients retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 12. The lines mark the respective 95% confidence interval. Coefficients that are (at least) significant on the 5% level are marked in blue.

Focusing on SAD_{t-1}, I find a large and statistically significant negative association in the upper tail: the coefficients at the 90th and 95th quantiles ($\tau = 0.90$ and $\tau = 0.95$) are negative at $p < 0.01$. In quantile-regression terms, conditional on field and year fixed effects and instrument dummies, higher relative subsystem costs in $t{-}1$ shift the upper conditional quantiles of percentage change downward. In short, when functional costs are high, extreme increases in policy output become less likely and less pronounced.

The estimated coefficients on SAD_{t-1} are -15 at $\tau = 0.90$ and -23 at $\tau = 0.95$. Interpreting SAD_{t-1} as a within-period, field-standardized measure (one-unit = one standard deviation), a one-unit increase shifts the 90th and 95th conditional quantiles of percentage change downward by 15 and 23 percentage points, respectively (conditional on field and year fixed effects and instrument dummies). These sizable shifts indicate a pronounced dampening effect of higher functional costs on extreme policy changes.

Because SAD_{t-1} can also take negative values (relatively low subsystem costs), the symmetry of the specification implies the reverse: a one–standard deviation decrease in SAD_{t-1} is associated with roughly $+15$ and $+23$ percentage point increases in the 90th and 95th conditional quantiles. When functional costs are low, the constraints on adjustment are weaker and the upper tail of the conditional distribution expands, hence, extreme changes become more pronounced.

Taken together, the estimates point to a directional association: higher subsystem costs are associated with a thinner right tail (smaller punctuations), whereas lower costs are associated with a wider right tail (larger shifts). This pattern is consistent with the theoretical expectation that functional costs relate to both the likelihood and the magnitude of extreme policy changes, in line with punctuated dynamics. This effect also holds for different specifications of the cost variable and in simpler bivariate models (see SI).

So far, I assumed that the costs affect different legal instruments similarly. Yet, it is plausible that the different instruments are differently affected by functional costs, since they vary in how far they have to interact with them. To check for this, I include an additional estimation with an interaction effect between the instrument dummies and SAD_{t-1}. For the exact specification see the SI. Figure 6 plots these interaction coefficients across quantiles, along with 95% confidence intervals. In the upper quantiles ($\tau = 0.90, 0.95$), the coefficient for regulations is positive and statistically significant ($p < 0.05$), indicating that subsystem costs dampen large year-to-year changes less for regulations than for decisions. The coefficient for directives is smaller and generally not statistically significant, suggesting no consistent difference from decisions.

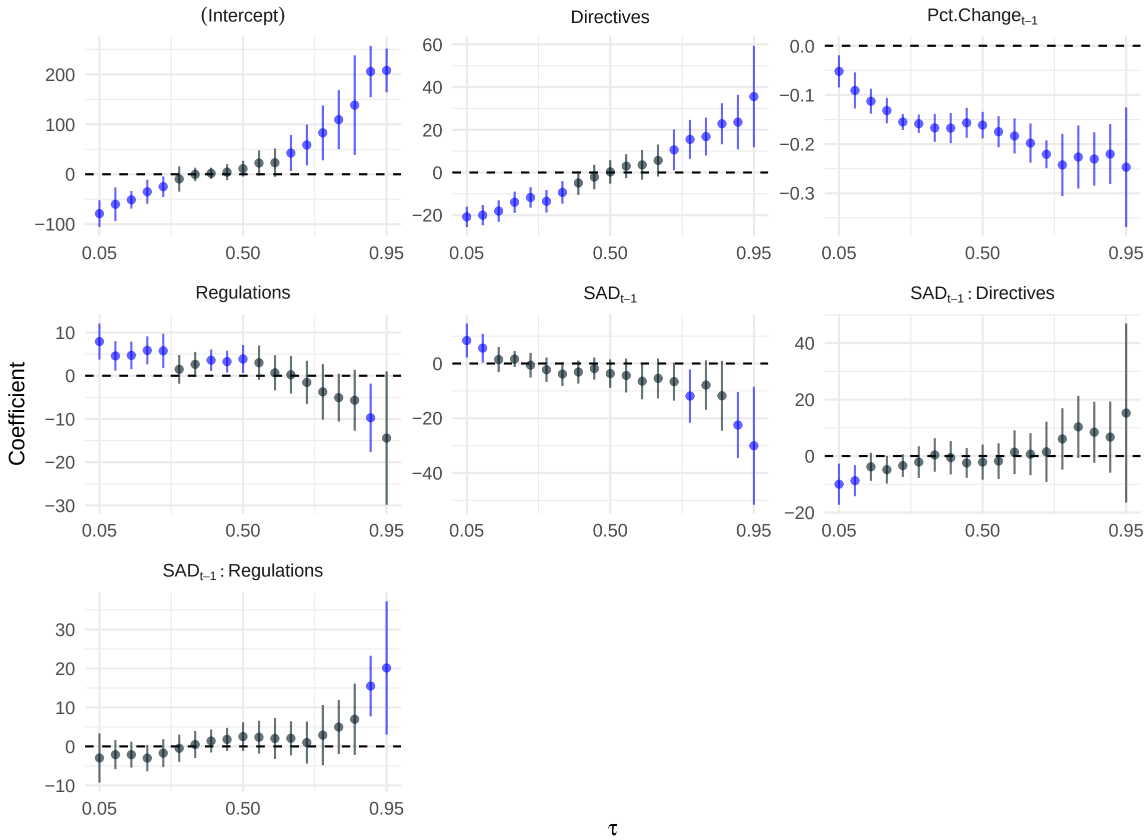

Figure 6: Interaction coefficients from quantile regression across multiple values of τ for the specification in Equation 14. Lines mark 95% confidence intervals; coefficients significant at the 5% level are marked in blue.

Therefore, the results suggest that not only do higher costs dampen the extreme upper tail, but this dampening is not uniform across legal instruments. As shown in Figure 6, regulations are less sensitive to functional costs than decisions, while directives exhibit no clear difference from the baseline. This empirical heterogeneity provides a natural motivation to extend the model. The baseline framework treated functional costs as exerting the same dampening force across all modes of law-making. Yet the evidence points to systematic differences: some instruments absorb costs more heavily at adoption, while others might buffer or defer them.

7 Model Extension: Heterogeneous Cost Sensitivity

To make sense of these differences, I now turn back to the model and extend it to incorporate heterogeneous cost sensitivity. So far, both the theoretical model and the main empirical analysis have assumed that functional costs exert a uniform effect on policy-making across all contexts. In the model's baseline utility setup (see Equation 5), the cost term enters identically for all policy outputs. A natural extension is to allow the effect of functional costs to vary across modes of law-making. In the extended utility formulation:

$$U_{t,i,j}^{(m)} = D_{t,i} - \kappa_m \, C_{t,i} - \sum_j P_{t,i,j} \, \phi_{t,i,j}^{(m)} \tag{13}$$

the multiplier κ_m scales how strongly mode m reacts to a given level of functional costs $C_{t,i}$.

As stated before, in complex policy systems the total functional costs $C_{t,i}$ are not a single burden but a bundle of drafting, harmonization, compliance, and implementation tasks. To clarify why κ_m differs across modes at the adoption stage (the stage captured by my dependent variable), it is useful to separate two dimensions:

1. Transmission at adoption: for each component $k \in \{\text{draft}, \text{harm}, \text{comp}\}$, let $\tau_{m,k} \in [0, 1]$ denote the share that must be absorbed now (cannot be deferred).
2. Processing discount: let $\psi_m \in (0, 1]$ summarize how allocation/capacity (centralization in expert venues) and modularity (scope for parallel work/standardization) buffer that work.

This yields a simple interpretive mapping:

$$\kappa_m \propto \psi_m \cdot (\tau_{m,\text{draft}} + \tau_{m,\text{harm}} + \tau_{m,\text{comp}}) \tag{14}$$

Here, *timing* (deferral vs immediacy) primarily shifts the τ's, while *allocation* and *modularity* shape ψ_m.

7.1 Legal instruments as an EU example

Linking this back to the empirical findings, the regression results suggest that subsystem costs do not constrain all instruments equally. In particular, regulations appear less sensitive

Table 1: Instrument mapping at adoption: transmission τ and processing ψ_m

Instrument	Transmission τ	Processing discount ψ_m	Net κ_m
Regulations	• τ_{draft}: *High* (single directly applicable EU text) • τ_{harm}: *Mod* (political core now; detail via delegated/implementing acts) • τ_{comp}: *Low* (direct applicability; admin build-out post-adoption)	• *Strong buffering*: centralized expertise (DGs, agencies, comitology) • Delegated/implementing-act pipelines; annex-based drafting $\Rightarrow$ parallelization/reuse	**Lower** κ_R
Directives	• τ_{draft}: *Mod–High* (general & transposable; definitions/options up front) • τ_{harm}: *Mod–High* (reconcile MS diversity ex ante: derogations, mutual-recognition) • τ_{comp}: *Very low* (formal compliance deferred; anticipatory design now)	• *Moderate buffering*: dispersed capacity (EU + 27 MS) • Modular completion externalized to MS; fewer EU-internal pipelines	$\kappa_{\text{Dir}} \approx \kappa_{\text{Dec}}$ (and $> \kappa_R$)
Decisions	• τ_{draft}: *High* (case-specific text) • τ_{harm}: *Low* (limited cross-sector reach) • τ_{comp}: *Low–Mod* (specific addressees; little EU-level deferral)	• *Weak buffering*: narrow scope; tightly integrated files • Minimal scope for delegation/modularity	**Baseline** κ_{Dec}

to costs than decisions, while directives show no consistent difference. Therefore, a plausible explanation is that EU legal instruments embody different configurations of transmission and processing capacity (Hurka and Haag 2020; Yordanova and Zhelyazkova 2019). These design features in turn determine how much of $C_{t,i}$ must be absorbed at adoption and how effectively that workload can be buffered. The more work can be deferred beyond adoption (lower τ), centralized in high-capacity venues, or modularized (lower ψ_m), the smaller the effective κ_m is likely to be at the adoption stage.

Viewed through this processing–transmission lens, the observed heterogeneity in the estimates aligns with institutional design as shown in a summary in Table 1. Regulations can defer part of the compliance workload and benefit from strong centralized pipelines, making them less sensitive to functional costs at adoption ($\kappa_R < \kappa_{\text{Dec}}$). Directives shift formal compliance beyond adoption but require heavy upfront harmonization and are buffered less effectively within EU venues, implying $\kappa_{\text{Dir}} \approx \kappa_{\text{Dec}}$. Decisions, which lack both modular deferral and centralized processing, remain closest to the baseline.

8 Discussion

This study highlights the critical role of functional costs in shaping policy punctuations within complex legislative systems, with the European Union (EU) serving as a case study.

The findings provide strong empirical support for the theoretical framework linking subsystem complexity, functional costs, and patterns of policy change. Specifically, functional costs—measured through the legal network's average degree—significantly influence policy output distributions, especially at the upper quantiles where extreme policy changes are observed.

A key insight is that the impact of functional costs varies by instrument. Relative to decisions, regulations exhibit weaker dampening at the top of the distribution: higher subsystem costs shrink extreme year-to-year changes less for regulations ($\kappa_{\mathrm{R}}.< \kappa_{\mathrm{Dec}}$), consistent with centralization and modularity buffering cost pressures at the adoption stage. By contrast, the directive interactions are small and not reliably different from decisions, suggesting that deferral to national transposition does not, at adoption, produce a strong buffering effect.

8.1 Practical Implications

These findings have important practical implications. Recognizing that functional costs are both influential and adaptable offers policy-makers a valuable lever for reform. By implementing institutional reforms, fostering innovation, or simplifying legislative processes, governments can reduce these costs over time. Although such strategies may require upfront investments and may not yield immediate results, the long-term benefits include reducing policy inertia and enabling more agile legislative action. Efforts to streamline drafting and compliance processes can prevent legislative stagnation. Early examples of this approach can already be seen in the EU's REFIT program, which aims to simplify legislation and reduce unnecessary burdens.

8.2 Limitations and Future Research

While these findings provide robust insights within the EU context, they may have limitations when generalized to other political systems. The EU's unique supranational structure, relative autonomy from direct electoral pressures, and complex regulatory framework differ significantly from national governments, where electoral cycles and public opinion often play a more direct role in shaping policy. However, the theoretical framework presented here could be adapted to other governance systems, as all policy-makers must navigate existing legal frameworks when developing new policies. The focus on functional costs and subsystem complexity offers a valuable foundation for exploring policy dynamics across diverse political environments.

One limitation of this study is the operationalization of functional costs through the average degree of the legal network, which may not capture all dimensions of these costs. While this metric reflects the complexity and interconnectedness of the legal subsystem, it may overlook factors such as administrative burdens, legal intricacies, or the actual resource expenditures involved in policy-making. Additionally, the assumption that network degree directly correlates with functional costs may not always hold, as some highly interconnected policy areas could mitigate additional costs through standardized procedures.

Moreover, measuring policy punctuations solely in terms of percentage changes might not fully capture the significance or substantive impact of these shifts. Percentage changes indicate the magnitude of change relative to a prior state but may not reflect the qualitative importance of those changes. Some policy developments may have far-reaching consequences despite only modest quantitative shifts, while large changes in less critical areas may have minimal overall impact. Future research could benefit from incorporating qualitative analyses and assessments of policy impact to provide a more comprehensive understanding of policy dynamics.

8.3 Conclusion

This paper has shown that functional costs—capturing the complexity and interdependence of legislative subsystems—play a decisive role in shaping the size and likelihood of extreme policy changes in the European Union. Using a dynamic feedback framework rooted in complex systems theory, I argued that these costs act as a form of negative feedback, dampening punctuations when they are high and loosening constraints when they are low. The empirical results align with this expectation: in the upper tail of the change distribution, higher relative subsystem costs are strongly and significantly associated with smaller punctuations, while lower costs correspond to larger shifts.

The analysis further demonstrates that this dampening effect is not uniform across legal instruments. Regulations, with their centralized drafting and direct applicability, show a notably weaker sensitivity to subsystem costs than decisions, consistent with the idea that centralized processing can buffer against cost-related constraints. Directives, by contrast, exhibit no consistent difference from decisions in their cost sensitivity at the EU adoption stage, suggesting that cost deferral to national implementation plays a more limited role than sometimes assumed. These patterns underscore that the interaction between functional costs and institutional design choices—particularly instrument choice—can shape how punctuated dynamics unfold in practice.

By integrating the concept of evolving, subsystem-specific functional costs into the study of policy punctuations, this study expands the explanatory scope of Punctuated Equilibrium Theory. It shows that not all "friction" is static and that cost structures themselves are part of the dynamic environment policy-makers navigate. The findings suggest that policy stability and instability emerge not only from shifts in attention or external shocks, but also from the ongoing accumulation, distribution, and management of functional constraints within complex policy systems.

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Supplementary Information

A.1 Density curves and Lorenz curve for Pct.changes in simulation

Figure A.1 shows the distribution of pct. changes derived from an exemplary simulation of the model presented in the main text.

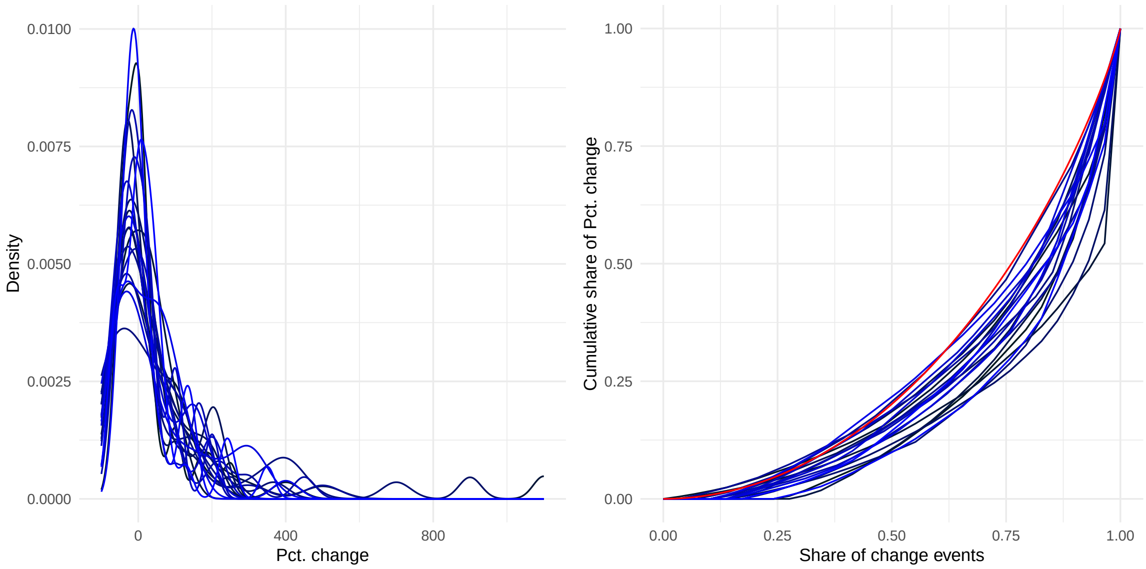


Figure A.1: Density and Lorenz curves of pct. changes of 20 simulated policy fields over a 30 year period

As the figure shows, all fields demonstrated punctuated patterns. To assess if the policy making across fields and instruments follows a punctuated pattern, I apply the method proposed by Kaplaner and Steinebach (2022). With the average Gini being $G = 0.551$ which implies a more punctuated distribution than a normal distribution ($G = .414$). Notably, this value lies close to the mean Gini value observed in the empirical distribution of EU legislative outputs (see below).

A.2 Density curves of Pct. changes by policy fields and legal instrument in the EU

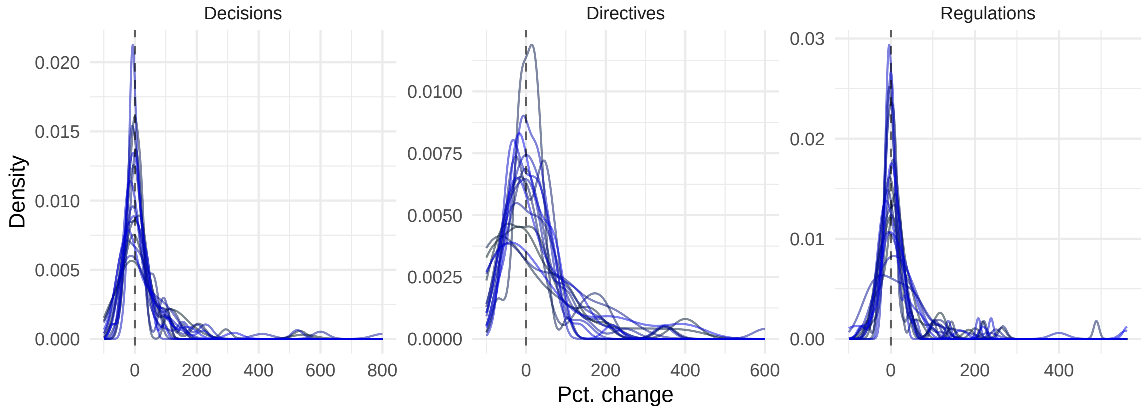


Figure A.2: Density curves of pct. changes across all policy fields split by legal instruments

A.3 Gini coefficient of Pct. changes by policy fields and legal instrument in the EU

Figure A.3 shows the Gini Coefficient based on the pct. changes of all policy fields split by legal instruments. The dotted line marks the Gini coefficient of a standard normal distribution (see above).

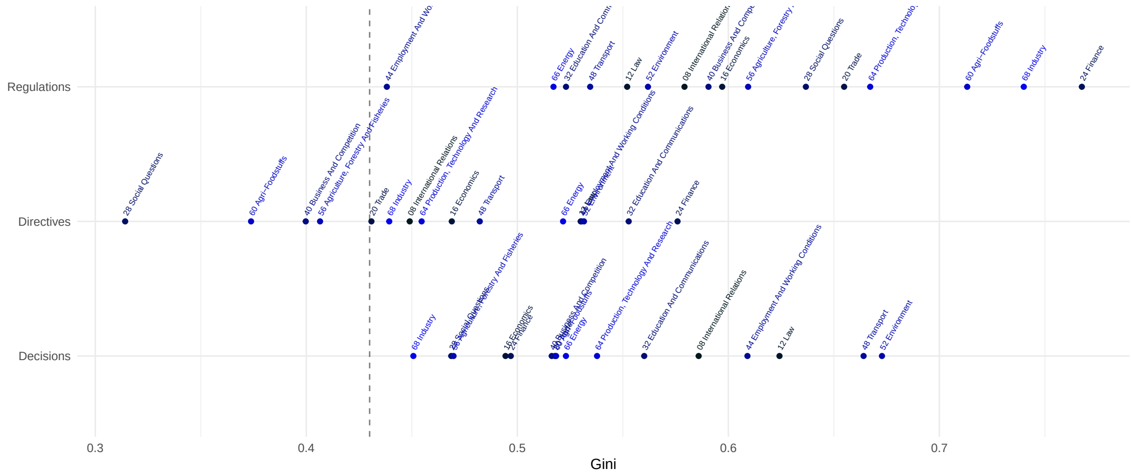
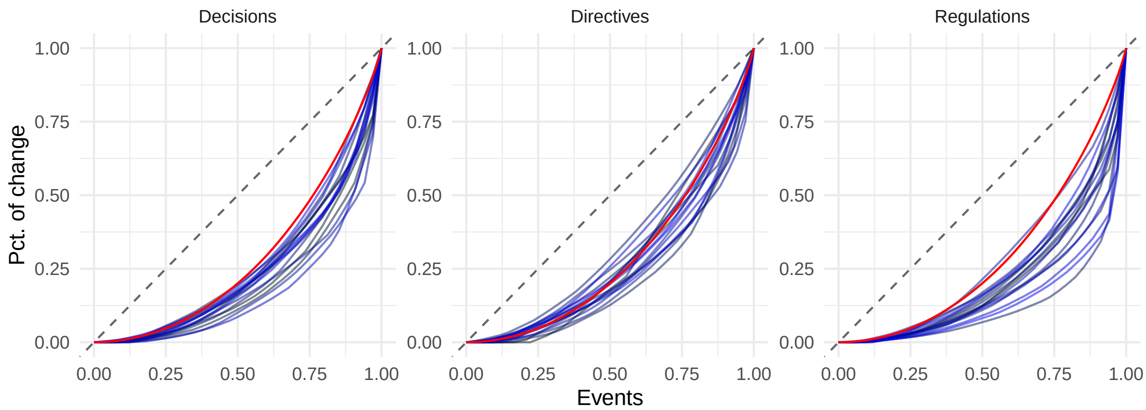


Figure A.3: Gini coefficient based on Pct. Changes per policy field and instrument, dotted line represents the Gini coefficient of a normal distribution. Values above that indicate a punctuated pattern (see Kaplaner and Steinebach 2022).

Looking at the graph, it is evident that nearly all fields across legal instrument have patterns that divert from a normal distribution.

A.4 Lorenz curves of Pct. changes by policy fields and legal instrument in the EU

The same pattern emerges when looking at the Lorenz-curve, that can visualize the inequality within distributions (Lorenz 1905). The red line indicates the Lorenz-curve of a normal distribution.



A.5 Main regression results table

Table A.1: Quantile regression results for selected quantiles.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-82.23*** [-108.35, -56.12]	9.75 [-8.09, 27.59]	206.4*** [175.11, 237.69]
SAD (lagged)	3.45 [-1.66, 8.56]	-3.62* [-7.86, 0.62]	-23.02*** [-40.06, -5.99]
Regulations	8.14*** [5.49, 10.78]	4.4*** [1.36, 7.44]	-9.05 [-21.73, 3.64]
Directives	-18.64*** [-21.62, -15.66]	1.11 [-4.73, 6.94]	32.49*** [19.63, 45.34]
Pct. change (lagged)	-0.06*** [-0.09, -0.02]	-0.16*** [-0.19, -0.14]	-0.23*** [-0.3, -0.16]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17116.35	17179.58	19874.81
BIC	17400.74	17463.97	20159.2

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6 Main regression results table with interaction

Table A.2: Quantile regression results for selected quantiles including interaction between SAD (lagged) and Legal instrument.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-78.81*** [-105.61, -52.02]	11.21 [-4.69, 27.1]	207.84*** [164.14, 251.54]
SAD (lagged)	8.39*** [2.17, 14.62]	-3.68 [-8.88, 1.53]	-30.06*** [-51.63, -8.49]
SAD (lagged):Regulations	-2.97 [-9.27, 3.33]	2.51 [-1.16, 6.19]	20.13** [3.02, 37.23]
SAD (lagged):Directives	-9.99*** [-17.26, -2.71]	-2.16 [-8.39, 4.07]	15.22 [-16.49, 46.93]
Regulations	7.92*** [3.75, 12.08]	3.87** [0.63, 7.12]	-14.41* [-29.85, 1.03]
Directives	-20.78*** [-25.53, -16.04]	0.27 [-5.22, 5.76]	35.55*** [11.77, 59.32]
Pct. change (lagged)	-0.05*** [-0.08, -0.02]	-0.16*** [-0.19, -0.13]	-0.25*** [-0.37, -0.13]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17103.4	17178.8	19862.05
BIC	17398.52	17473.92	20157.17

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.7 Descriptive table main variables

Table A.3: Descriptive statistic main variables by policy field.

		Pct. Change at quantile:			
Field	N	5th	50th	95th	Mean SAD
08 International Relations	83	-64.1	1.64	145	-0.26
12 Law	93	-70.1	8.7	185	-0.909
16 Economics	89	-100	0	160	1.42
20 Trade	99	-29.7	1.71	60.3	0.767
24 Finance	98	-51.7	0	176	-0.637
28 Social Questions	99	-41.2	6.67	78.3	0.39
32 Education And Communications	99	-50	6.25	126	-0.954
40 Business And Competition	96	-68.2	0	175	1.81
44 Employment And Working Conditions	98	-57.9	0	154	-0.534
48 Transport	99	-59.2	4.17	145	-0.21
52 Environment	99	-55.6	3.03	157	0.236
56 Agriculture, Forestry And Fisheries	99	-39.6	1.07	62.1	0.732
60 Agri-Foodstuffs	99	-41.3	-0.411	111	0.206
64 Production, Technology And Research	99	-50	1.67	138	-0.843
66 Energy	91	-78.4	0	200	-1.23
68 Industry	99	-42.2	2.86	75.8	-0.0059

A.8 Robustness to different specification of independent variable

To assess the robustness of the model to the construction of the main independent variable, SAD, I apply an alternative transformation. Instead of scaling by period, I use ranked values. The results remain robust under this adjustment, demonstrating the model's stability. Only in the model including the interaction effect the significance level drops slightly for the 95th quantile, only being significant on the 10% level.

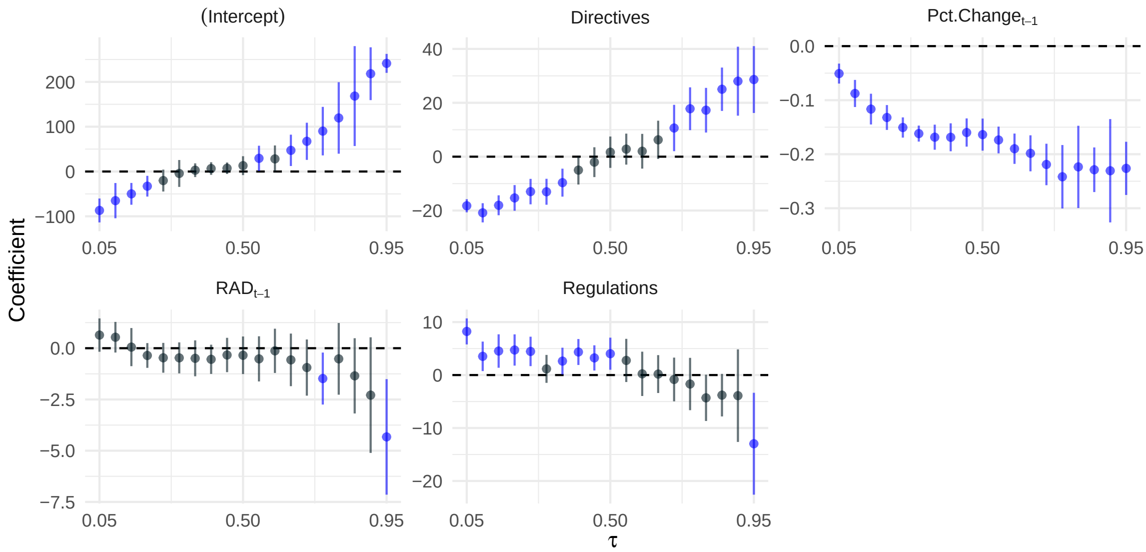


Figure A.4: Coefficient retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 12 with different dependent variable (ranked, RAD_{t-1}). The lines mark the respective 95% confidence interval. Coefficient that are (at least) significant on the 5% level are marked in blue.

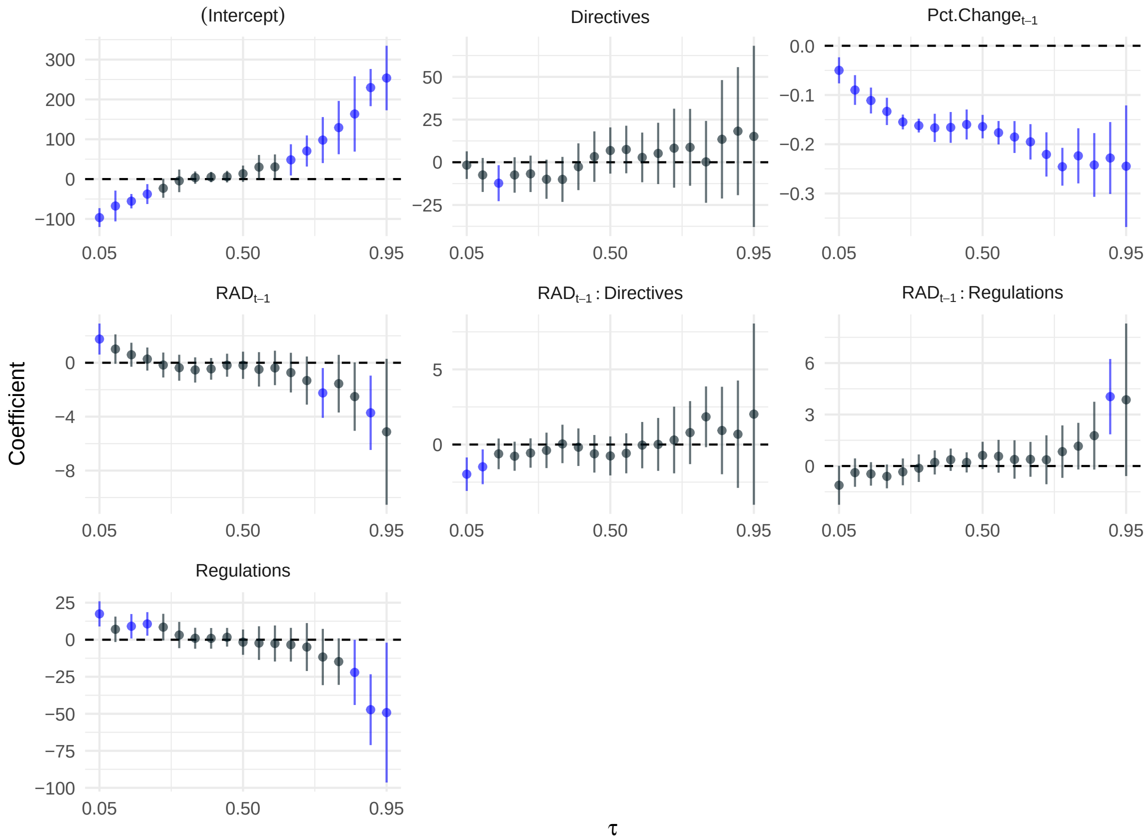


Figure A.5: Coefficient retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 14 with different dependent variable (ranked, RAD_{t-1}). The lines mark the respective 95% confidence interval. Coefficient that are (at least) significant on the 5% level are marked in blue.

A.9 Bivariate regression using main variable SAD

To evaluate potential suppression effects from the control variables, I estimate a simplified bivariate model that includes only the main independent variable, SAD, excluding all controls. As illustrated in Figure A.6, the effect persists even in this reduced model. The results indicate that the effect remains similar in magnitude to the main model (-20% compared to -23%) and is statistically significant at the 10% level.

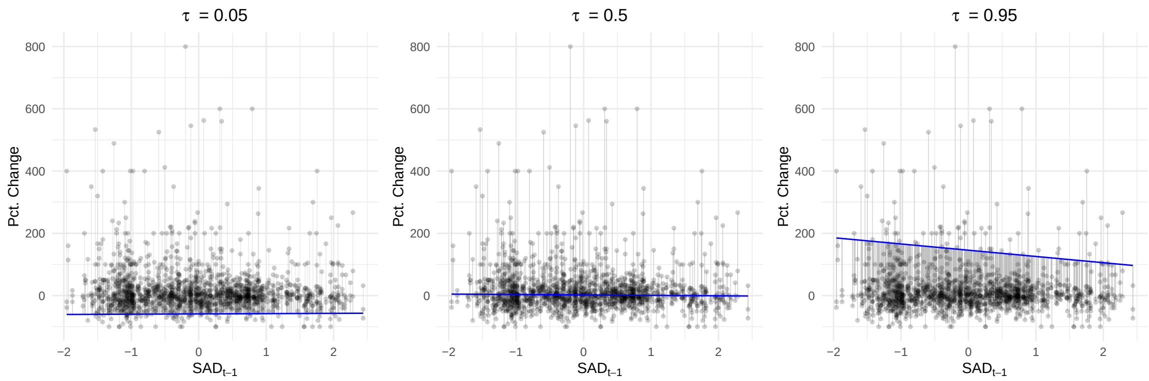


Figure A.6: SAD_{t-1} on x-axis, Pct. Change on y-axis. Blue line indicates regression line for different quantiles. Grey lines indicate distance from regression line.

Table A.4: Quantile regression results for selected quantiles for bivariate regression without controls.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-58.94*** [-63.3, -54.57]	2.05** [0.33, 3.76]	145.73*** [127.35, 164.11]
SAD (lagged)	0.9 [-3.69, 5.49]	-1.38 [-3.02, 0.26]	-20.08* [-38.89, -1.27]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17734.15	17327.1	20762.93
BIC	17744.88	17337.83	20773.66

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.10 Empirical specification of instrument heterogeneity

To examine whether cost sensitivity varies systematically by legal instrument, I estimate the same quantile regression model as before in which SAD_{t-1} is additionally interacted with the legal instrument dummies, using decisions as the reference category:

$$\begin{aligned} Q_\tau(\text{Pct. change}_{i,l,t} \mid X) = {} & \alpha_\tau + \beta_{1,\tau}\,\text{Pct. change}_{i,l,t-1} + \beta_{2,\tau}\,\text{SAD}_{i,t-1} \\ & + \gamma_\tau^R\,R_l + \gamma_\tau^D\,D_l \\ & + \delta_\tau^R\,\text{SAD}_{i,t-1}\cdot R_l + \delta_\tau^D\,\text{SAD}_{i,t-1}\cdot D_l \\ & + \lambda_{i,\tau} + \mu_{t,\tau}\,. \end{aligned} \tag{15}$$

The interaction coefficients δ_τ^R and δ_τ^D capture the change in the estimated cost multiplier κ_m for regulations and directives relative to decisions.